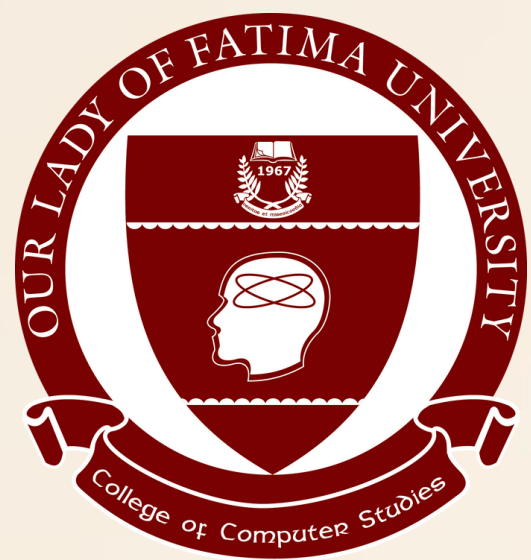




Module 1

Computing Devices

ITCL112 - INTRODUCTION TO COMPUTING W/ LAB

INTRODUCTION



The computing industry is constantly evolving, and new types of computing devices are being developed all the time. As technology advances, computing devices will become even more powerful and versatile, and they will play an even greater role in our lives.

MODULE 1

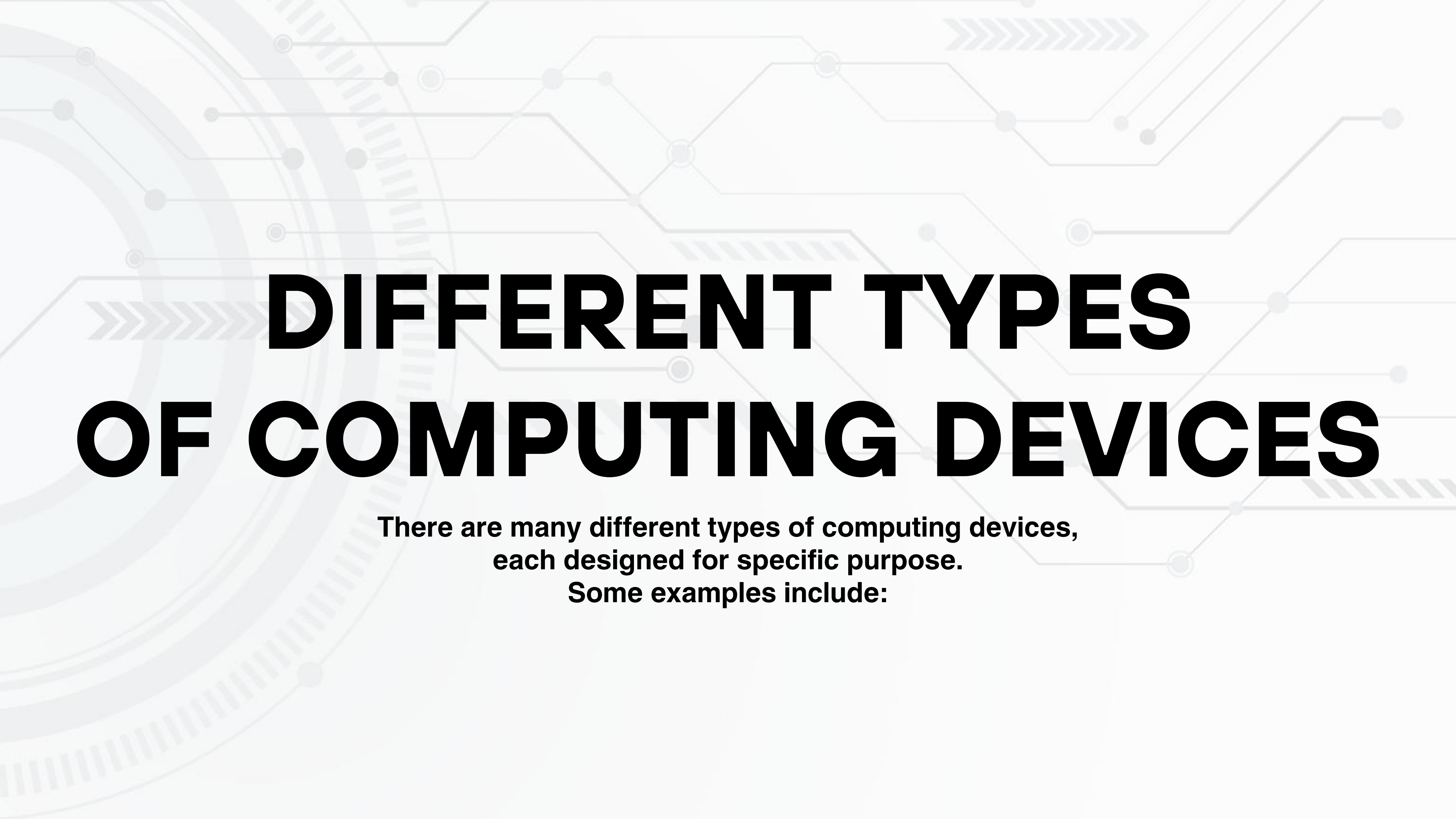
- 1. What is computing devices**
- 2. Different types of computing devices**



What is Computing Devices?

A computing device is a machine that can be programmed to carry out sequences of arithmetic or logical operations (computation) automatically. Modern digital electronic computers can perform generic sets of operations known as programs. These programs enable computers to perform a wide range of tasks.

A computing device can consist of a standalone unit or several interconnected units. It can also be a device that provides a specific set of functions, such as a phone or a personal organizer, or more general functions such as a laptop or desktop computer.

The background of the slide features a complex, futuristic circuit design. It includes various geometric shapes like circles, lines, and polygons, some of which are highlighted with a light blue glow. There are also several sets of parallel lines that resemble circuit traces or data paths. The overall color scheme is light gray and white, with the text in bold black.

DIFFERENT TYPES OF COMPUTING DEVICES

**There are many different types of computing devices,
each designed for specific purpose.
Some examples include:**

Personal Computer

A personal computer (PC) is a multi-purpose microcomputer whose size, capabilities, and price make it feasible for individual use. Personal computers are intended to be operated directly by an end user, rather than by a computer expert or technician.

Characteristics of a Personal Computer:

Single-user focus

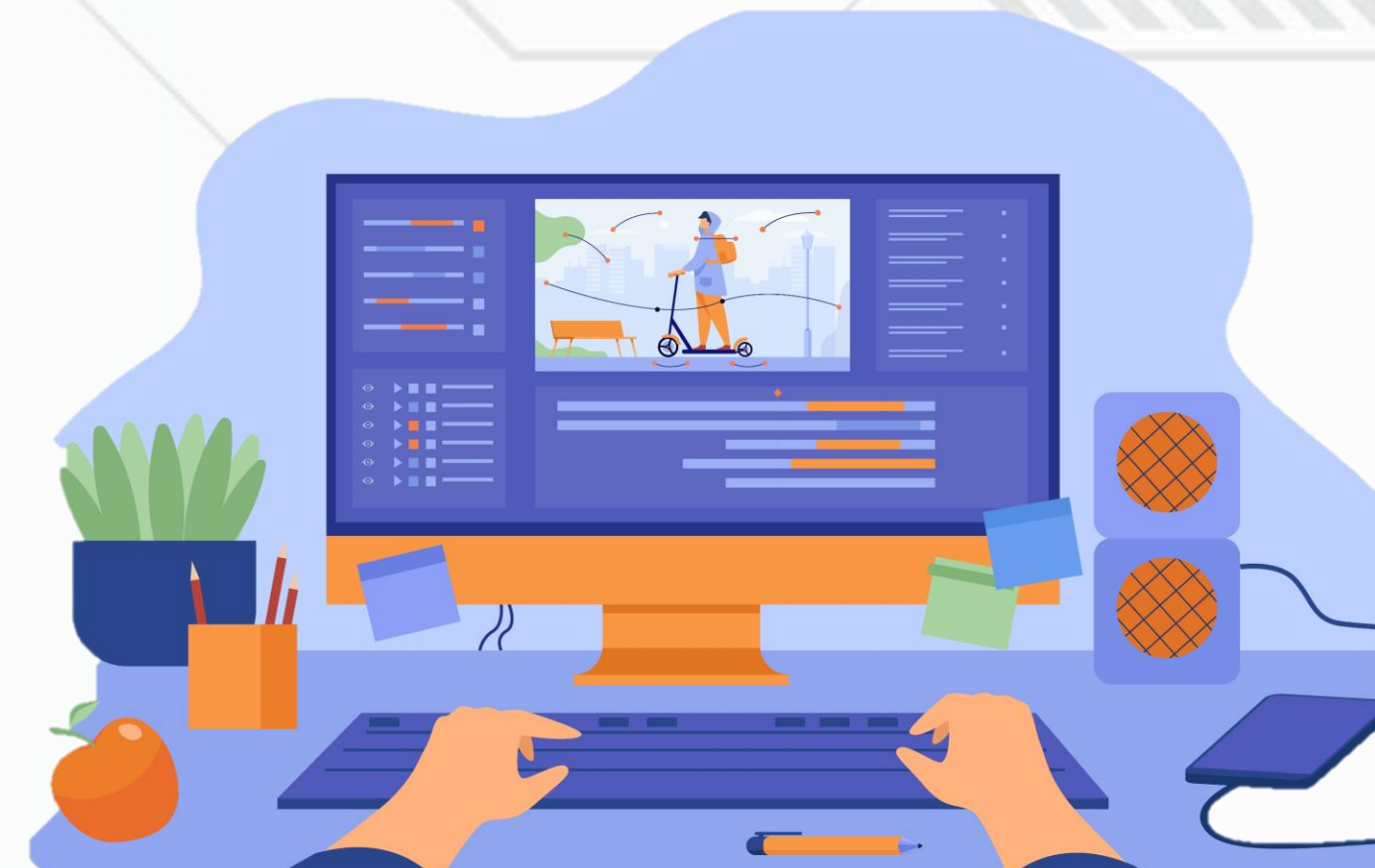
Affordable

User-friendly

Versatile

Compact size

a personal computer is the computer you typically see at homes, schools, and offices built for personal and everyday use.



Desktop Computer

A desktop computer is a type of personal computer (PC) that is designed to be used at a single, fixed location such as a desk or workstation. Unlike laptops or mobile devices, desktop computers are not portable and usually consist of several separate components.

Key Features of a Desktop Computer:

Main Unit (CPU Tower/Case)

Monitor

Keyboard and Mouse

Power Supply

Expandable and Upgradable



Notebook Computer

A notebook computer is a small, portable personal computer that is designed for mobility and convenience. It is commonly known today as a laptop.

Key Characteristics of a Notebook Computer:

Portable:

Built-in
Components:

Battery
Powered:

Wireless
Connectivity:

All-in-One
Design:



All in One Computer

An All-in-One (AIO) computer is a type of computer where all the main components such as the CPU (processor), memory, storage, and motherboard are built into the same case as the monitor. Unlike traditional desktop computers, which have a separate tower and monitor, an AIO combines everything into one sleek unit.

Best For: Home use, office work, education, and users who prefer a clean, minimalist setup.

Example:

- Apple iMac
- Microsoft Surface Studio
- HP Pavilion All-in-One
- Lenovo IdeaCentre AIO



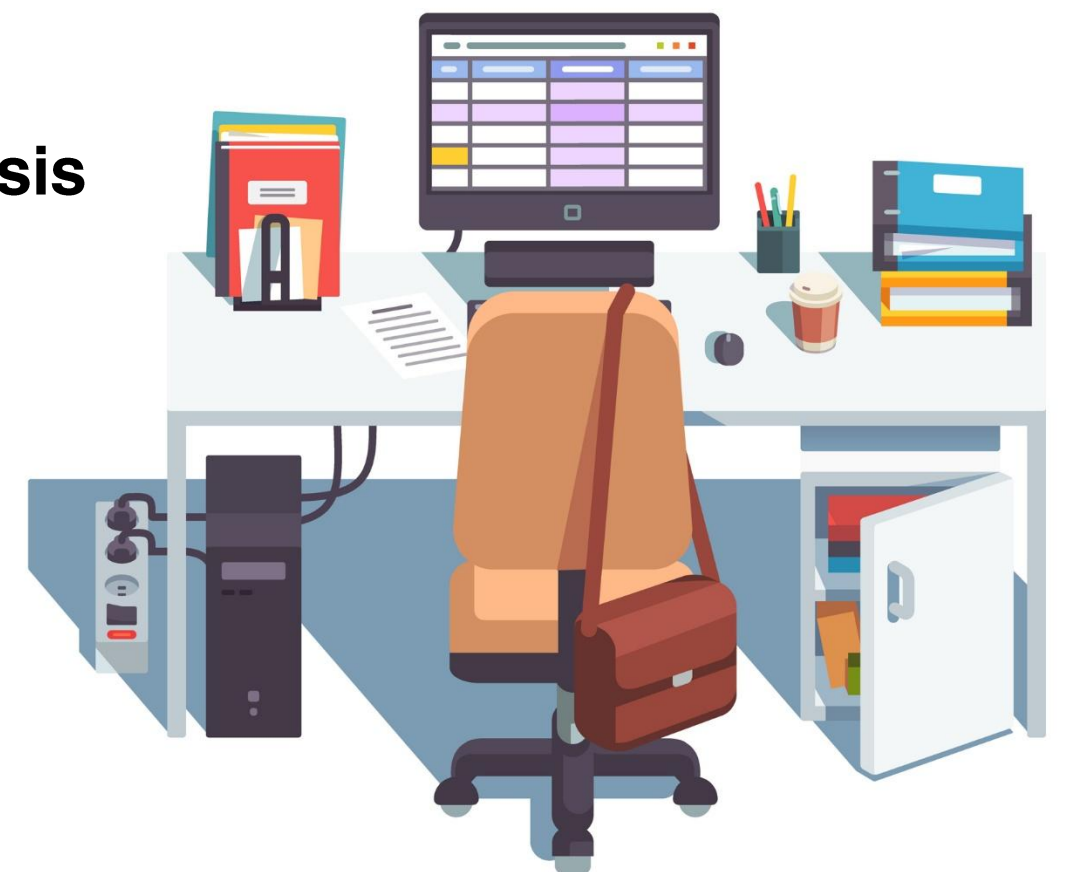
Workstation Computer

A workstation computer is a high-performance desktop computer designed for technical or scientific applications that require more power, reliability, and capabilities than a standard personal computer.

A workstation is like a professional-grade computer more powerful than a regular desktop, but not as large or complex as a server.

Example:

- **3D modeling and animation**
- **Engineering design (AutoCAD, SolidWorks)**
- **Scientific simulations**
- **Video editing and production**
- **Machine learning and data analysis**



Tablet Computer

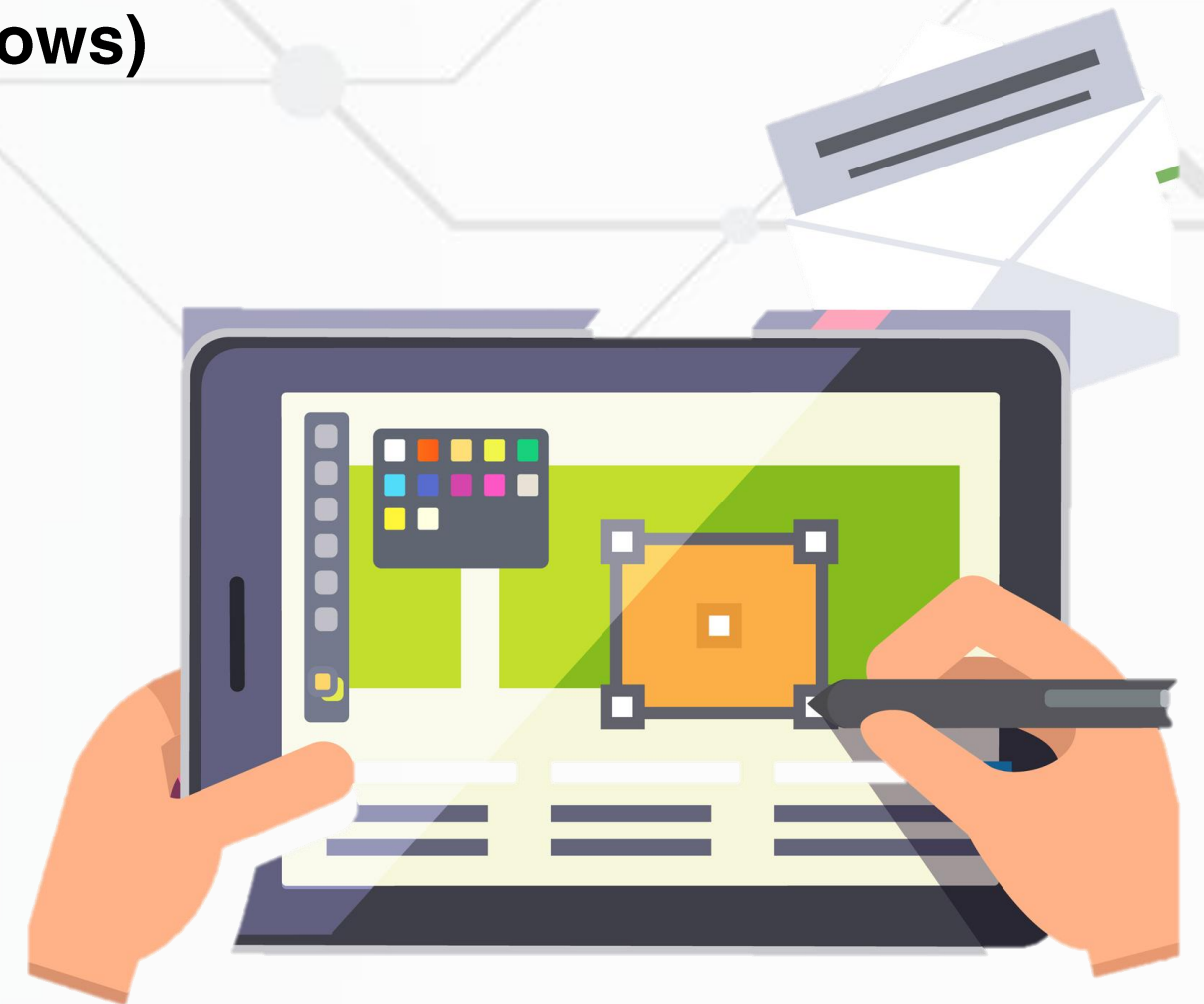
A tablet computer (or simply tablet) is a portable, wireless, touch-screen device that combines features of a smartphone and a laptop. It is typically larger than a smartphone but smaller and lighter than a laptop.

A tablet computer is like a bigger smartphone with better performance and a lighter laptop alternative for browsing, watching videos, reading e-books, studying, or working.

Examples:

Apple iPad (iPadOS), Samsung Galaxy Tab (Android)

Microsoft Surface Go/Pro (Windows)



Hand-Held Computer

A Handheld Computer (sometimes called a palmtop computer or PDA Personal Digital Assistant) is a small, portable computing device that you can hold in one hand and operate with the other.

A handheld computer is a mini-computer designed to fit in your hand, combining computing, communication, and personal organization features.

Examples: Early PDAs (Palm Pilot, HP Jornada), Pocket PCs, and modern smartphones (which evolved from handheld computers).

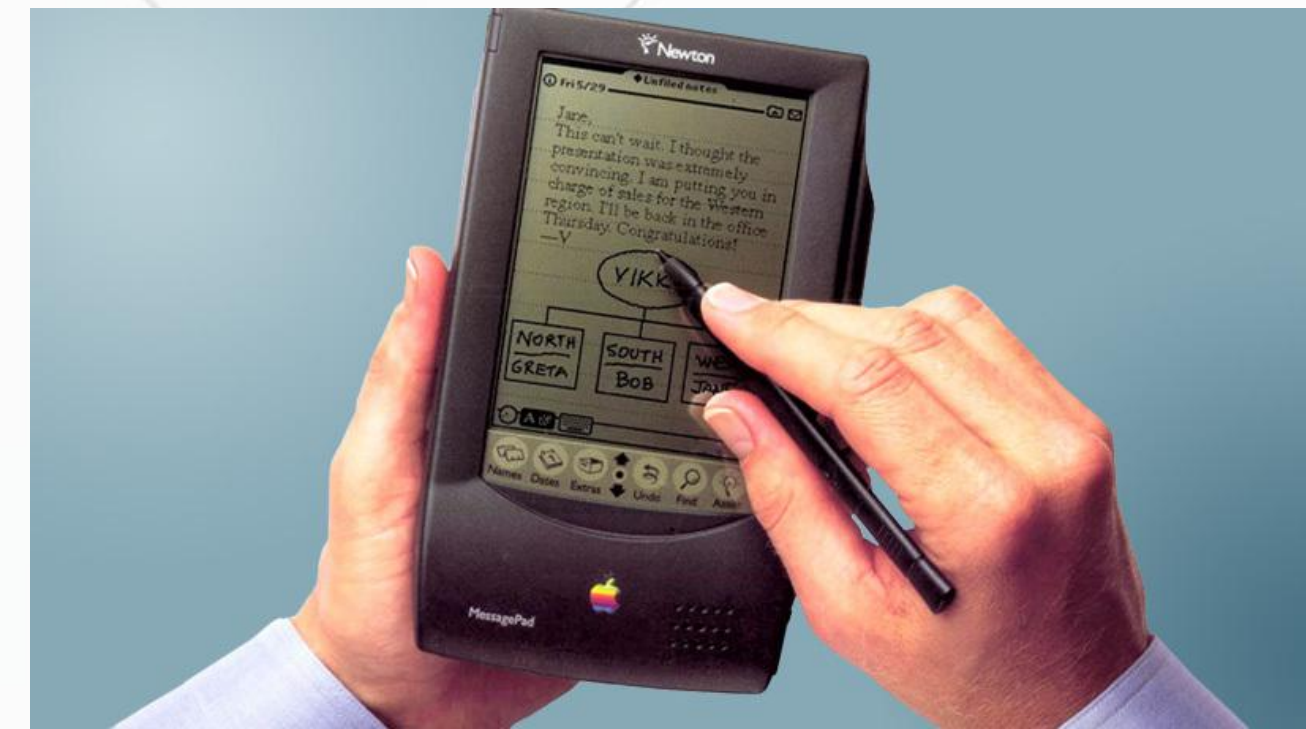


Personal Digital Assistants (PDA)

A Personal Digital Assistant (PDA) is a handheld electronic device that functions as a personal information manager. PDAs were popular before smartphones became mainstream.

Over time, PDAs evolved and merged with mobile phones, eventually becoming what we now know as smartphones.

Examples of PDAs: Palm Pilot, HP iPAQ, and early Windows Pocket PCs.



Smartphone

A smartphone computer (commonly just called a smartphone) is a handheld mobile device that functions like both a phone and a computer. It combines features of a traditional cell phone (calling, texting) with the processing power and capabilities of a computer

Smartphones run advanced OS like Android or iOS that allow multitasking and app installations. Equipped with processors, memory (RAM), storage, cameras, and sensors, similar to what computers have but in miniaturized form.



E-Book Reader Devices

An E-Book Reader Device (often called an e-reader) is a portable electronic device designed primarily for reading digital books, magazines, newspapers, and other text-based content. Unlike tablets or smartphones, e-readers are optimized for reading, often using special display technology to mimic the look of paper.

Most e-readers (like Amazon Kindle, Kobo, Nook) use electronic ink (E-Ink) screens that look like printed paper, reduce eye strain, and can be read comfortably even in bright sunlight.

Examples of Popular E-Book Reader Devices:

- Amazon Kindle series (Kindle, Kindle Paperwhite, Kindle Oasis)
- Kobo eReaders
- Barnes & Noble Nook
- PocketBook Reader
- Onyx BOOX



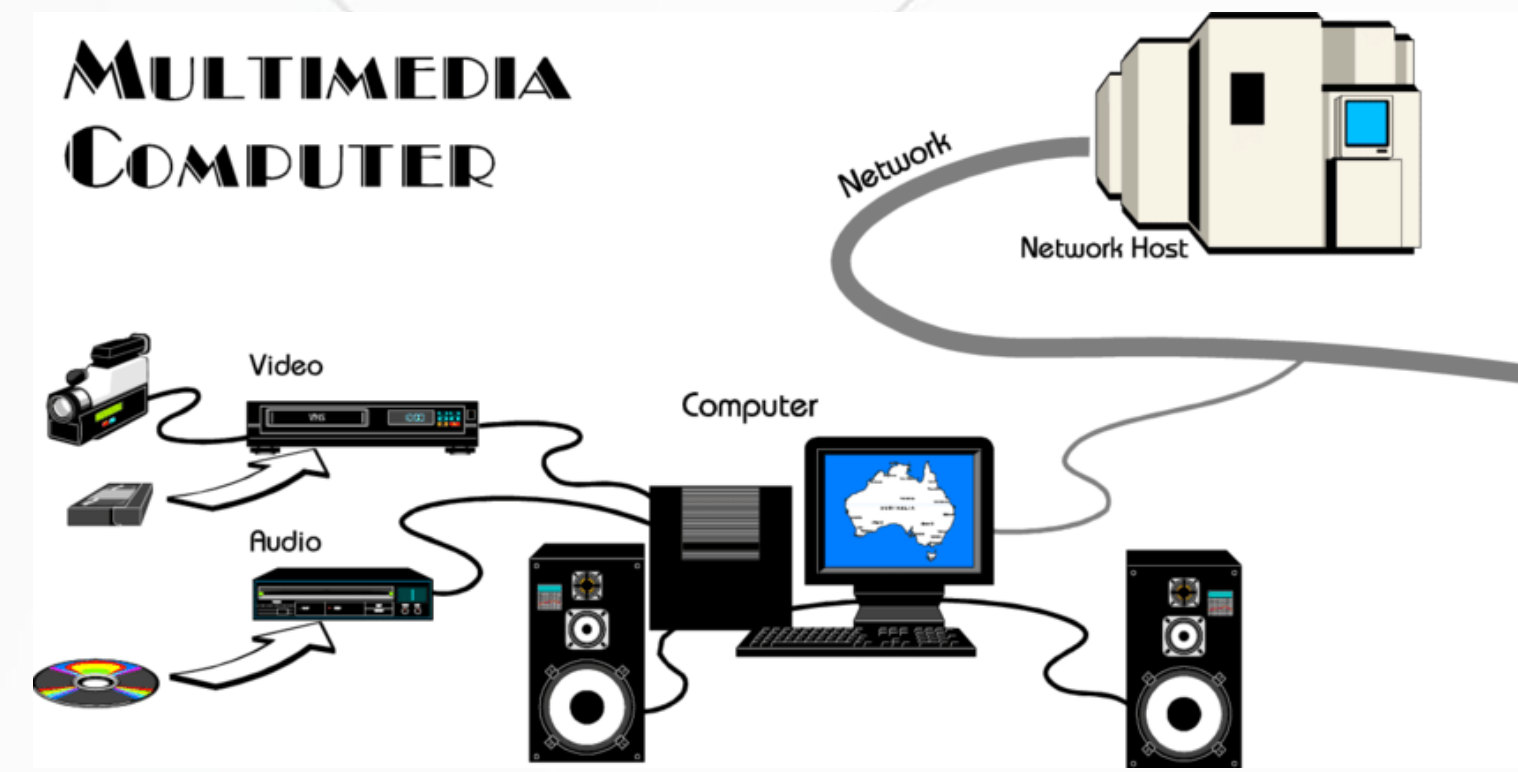
Multimedia Computers

Multimedia Computer is a computer system that is specifically designed to handle multimedia data such as text, images, audio, video, animation, and graphics in an integrated way.

a multimedia computer is a powerful system designed to create, process, store, and present multimedia content effectively.

Uses of Multimedia Computers:

- **Education - Interactive learning, e-books, simulations.**
- **Entertainment - Gaming, movies, music, streaming.**
- **Business - Presentations, advertisements, virtual meetings.**
- **Design & Creativity - Graphic design, animation, video editing.**
- **Communication - Video conferencing, online collaboration.**



Super Computer

A supercomputer is a very powerful computer designed to perform extremely complex and large-scale calculations at very high speeds. Unlike regular personal computers or even servers, supercomputers are built to handle massive amounts of data and solve problems that require enormous computational power.

High Processing Speed – Measured in FLOPS (floating point operations per second), often in petaflops or even exaflops. They use thousands or even millions of processors working together simultaneously.

Uses of Supercomputers:

- **Scientific Research** - Climate modeling, astronomy, physics simulations.
- **Medicine** - Drug discovery, genomics, disease modeling.
- **Engineering** - Aircraft and car design, nuclear research.
- **National Security** - Cryptography, defense simulations.
- **Artificial Intelligence** - Training very large AI models.



Super Computer

Mainframe Computer

A mainframe computer is a large, powerful, and high-performance computer system designed to handle massive amounts of data processing and support thousands of users simultaneously. It is mainly used by large organizations such as banks, insurance companies, government agencies, and airlines for critical applications like transaction processing, enterprise resource planning, and bulk data storage.

A mainframe computer is not something used at home or in small offices it's a business-critical machine for organizations that need speed, reliability, and the ability to process massive data continuously.

Examples of Use:

- **Banking** - Processing millions of ATM, credit card, and online transactions daily.
- **Airlines** - Managing reservations, ticketing, and scheduling.
- **Government** - Handling census, taxation, and national records.

